

AI & API Hackathon 2026 : First Edition

1. The Story

Every day, thousands of public infrastructure issues go unnoticed or remain unresolved. A pothole on a busy road, a broken streetlight, a leaking water pipeline, or illegally dumped waste may seem like isolated incidents, but together they affect public safety, transportation, sanitation, and the quality of life for millions of citizens.

Although citizens frequently report these problems through social media, phone calls, emails, or local offices, the process is often fragmented and inefficient. Reports are duplicated, descriptions are incomplete, locations are unclear, and government departments spend valuable time manually reviewing, categorizing, and forwarding complaints before any action can begin.

To address this challenge, the Ministry of Civic Infrastructure has decided to modernize the reporting process through an AI-powered digital platform that connects citizens and government agencies within a single ecosystem. The platform should intelligently analyze submitted reports, estimate their severity, identify possible duplicate submissions, and assist government officials in prioritizing and managing cases more effectively.

Your team has been contracted to design and build the first functional prototype of this platform during the hackathon. The solution should demonstrate how artificial intelligence, modern web technologies, and external APIs can improve public service delivery while creating a seamless experience for both citizens and government officials.

The final system should not simply collect reports. It should help authorities make faster, more informed decisions by transforming unstructured citizen submissions into structured, actionable information that can be tracked from initial reporting to final resolution.

2. The Problem Statement

Build a full-stack, AI-powered web platform that enables citizens to report public infrastructure issues while providing government authorities with the tools needed to review, prioritize, manage, and resolve those reports efficiently.

The platform should transform unstructured citizen submissions into structured, actionable information using artificial intelligence and meaningful external API integrations. Rather than acting as a simple complaint collection system, the application should assist decision-makers by automatically analyzing reports, estimating their severity, identifying possible duplicate submissions, and presenting information in a way that supports faster and more effective response.

The application must provide two primary user experiences:

- A **Citizen Portal**, where members of the public can submit infrastructure issues and track the progress of their reports.
- A **Government Management Portal**, where authorized officials can review reports, assign responsibilities, monitor progress, and update resolution status.

Throughout the application, AI should play a meaningful role in improving the quality and usefulness of submitted reports. Teams are encouraged to demonstrate thoughtful AI integration rather than using AI as a standalone feature.

The completed solution should demonstrate strong software engineering principles, including clean architecture, reliable data management, responsive user experience, secure handling of information, and appropriate use of external services. Judges will evaluate not only the final product but also the quality of its implementation, usability, and technical design.

The objective is to build a realistic prototype that demonstrates how AI and modern web technologies can improve civic services and public infrastructure management in real-world environments.

Optional Extension — Beyond the Core Requirements

Teams that successfully complete the core requirements may extend their solution with additional intelligent capabilities that provide meaningful value to citizens or government officials. These enhancements may include advanced AI features, geospatial visualization, multilingual interaction, predictive analytics, real-time collaboration, image understanding, or other innovative functionalities that strengthen the overall platform. Optional extensions are intended to encourage creativity and technical excellence without affecting the evaluation of the mandatory core requirements. Outstanding implementations may receive bonus consideration from the judging panel.

3. What You Need to Build

Your solution should include the following core components.

Component	Description
Citizen Reporting Portal	A user-friendly interface that allows citizens to report infrastructure issues, provide relevant details, upload supporting evidence (optional), and receive a unique tracking code.
Government Management Dashboard	A secure dashboard for reviewing reports, managing their lifecycle, assigning responsible departments, and monitoring platform activity.
AI-Powered Report Analysis	Analyze submitted reports using AI to categorize issues, generate structured summaries, estimate severity, and produce meaningful insights.
Duplicate Detection System	Detects potentially duplicate reports using appropriate similarity techniques while preserving all submitted reports.
Progress Tracking System	Allow citizens to track the status, progress history, and assigned department of their reports using a tracking code.
Persistent Data Storage	Store reports, AI-generated outputs, status history, department assignments, and other application data in a persistent database.
External Service Integration	Meaningfully integrate one or more external services, such as AI APIs, mapping services, cloud storage, or notification systems.

Responsive User Experience	Build a modern, intuitive, and responsive web application that works seamlessly across different screen sizes and devices.
Optional Innovation	Extend the platform with additional features that demonstrate creativity, technical excellence, and real-world value.

Technical Freedom

Teams are free to choose any programming language, framework, database, AI model, SDK, cloud platform, or software architecture. The judging panel will evaluate the quality, functionality, and engineering of the final solution rather than the technologies used.

4. Functional Requirements

Module 1 — Citizen Report Submission

Develop an efficient citizen-facing interface to submit infrastructure issues.

- **Submission Features:** Issue description, location, optional contact details, optional photo/URL evidence.
- **Output:** Generates a unique report ID and public tracking code.
- **Categorization:** Supports categories like *Pothole*, *Broken Streetlight*, *Water Leak*, *Illegal Dumping*, and *Other*. Citizens may select one, but the AI system generates or validates the final category.

Module 2 — AI Report Analysis

AI must process every submission to assist officials with efficient review.

- **Tasks:** Analyzes description, generates an issue category, creates a concise summary, and assigns a confidence score.
- **Output:** Returns validated, structured data prior to storage.

Module 3 — Severity Assessment

Evaluates report priority based on public safety risks, service impact, scale, immediate danger, or proximity to sensitive areas (schools, hospitals, main roads).

- **Output:** Generates a Severity Level, Severity Score, and brief rationale to drive dashboard prioritization.

Module 4 — Duplicate Detection

Intelligently flags potential duplicate reports without blocking new submissions.

- **Factors:** Compares location/coordinates, AI category, semantic description similarity, timing, and optional image similarity.
- **Behavior:** Keeps potential duplicates accessible while linking them to the primary report.

Module 5 — Government Management Dashboard

A centralized hub for officials to manage workflows and monitor metrics.

- **Core Actions:** View all reports; review AI analysis; search by keyword, location, or ID/code; filter by category, severity, status, or department.
- **Workflow:** Assign to departments, update status, add progress notes, and view operational analytics.

Module 6 — Public Progress Tracking

Allows citizens to monitor their report using their public tracking code.

- **Displays:** Summary, AI category, severity level, status, assigned department, submission date, progress history, and public updates.
- **Privacy:** Excludes sensitive personal information.

Module 7 — Data Management & External Integrations

- **Data Storage:** Uses persistent storage to retain reports, AI analysis, progress logs, department assignments, duplicate links, and system metadata.
- **Integrations:** Must meaningfully integrate at least one external service (e.g., AI APIs, Mapping/Geocoding, Cloud Storage, SMS/Email, or Environmental Data). Passive or trivial API calls do not qualify.

Minimum Required Screens

The completed application should include, at minimum:

Screen	Purpose
Citizen Report Submission	Submit new infrastructure reports
Submission Success Page	Display report confirmation and tracking code
Public Tracking Page	Track report progress using the tracking code
Government Dashboard	Review, manage, and monitor reports
Report Details & Management	View complete report information and update its lifecycle

6. Technical Expectations & Constraints

The completed solution should demonstrate sound software engineering practices, maintainable architecture, and appropriate use of modern web technologies. While teams are free to choose their preferred technology stack, the application should satisfy the following technical expectations.

Technical Expectations

Area	Expectation
Architecture	Design a clear and maintainable frontend and backend architecture with proper separation of concerns.
Input Validation	Validate user inputs on both the client and server sides to ensure data integrity.
Database	Store application data using a persistent database. In-memory storage alone will not receive full marks.
API Design	Implement clean, structured, and consistent API responses with appropriate HTTP status codes.

AI Integration	AI should provide meaningful value to the platform rather than serving as a superficial feature.
Error Handling	Gracefully handle validation failures, API errors, AI failures, database exceptions, and unexpected system errors.
Security	Protect sensitive information and securely manage API keys, secrets, and environment variables.
User Experience	Provide responsive layouts, loading indicators, and informative error or success messages.
Code Quality	Write clean, modular, readable, and maintainable code with appropriate documentation where necessary.

Constraints

- The solution must be developed during the on-site hackathon.
- The application must be a functional web-based product.
- Core application logic should be developed by the participating team.
- AI-assisted development tools are permitted.
- Third-party libraries and APIs may be used but should be appropriately credited in the project documentation.
- The application should remain functional even if optional features are not implemented.

7. Evaluation Criteria

The judging panel will evaluate each submission based on functionality, technical implementation, engineering quality, and overall user experience. Teams should prioritize completing the core requirements before implementing optional enhancements.

Evaluation Area	Marks
Citizen Reporting Experience	10
Government Dashboard & Report Management	15

Progress Tracking & Status History	15
AI Categorization & Structured Output	15
Severity Scoring Quality & Explainability	10
Duplicate Detection Approach	15
Database Design, Validation & Error Handling	10
UI/UX Quality & Responsiveness	5
Code Quality & Documentation	5
Total	100

Bonus Marks

Teams that successfully complete the core requirements may earn additional bonus marks by implementing innovative features that enhance the platform's functionality and user experience.

Bonus Feature
Interactive map-based issue visualization
Functional deployment (publicly accessible)
Image-based issue analysis using AI
Real-time dashboard updates
Multilingual support (Bangla & English)
Smart department recommendation
AI-generated resolution suggestions
Push notifications or email/SMS updates
Offline-first/PWA implementation
Any other innovative feature with real-world value

Bonus features will only be considered after the core requirements have been successfully implemented. Additional features cannot compensate for missing mandatory functionality.

Judging Considerations

Beyond the allocated marks, judges may consider the following qualities when distinguishing submissions of similar technical merit:

- Completeness of the solution
- Real-world applicability
- Creativity and innovation
- Quality of AI integration
- Software architecture and engineering practices
- Demonstration quality and presentation
- Team's understanding of design decisions and technical trade-offs

Tie-Breaking

If two or more teams achieve the same total score, the judging panel may consider one or more of the following:

- Quality and effectiveness of AI integration
- Overall system architecture
- Code quality and maintainability
- Innovation beyond the minimum requirements
- Live demonstration and judges' Q&A performance

8. Hackathon Rules, Submission & Deliverables

Hackathon Rules

To ensure a fair and competitive environment, all participating teams must comply with the following rules throughout the competition.

Rule	Description
Project Repository	A new public GitHub repository must be created during the on-site hackathon.
Pre-written Code	Pre-written source code, completed projects, or reusable project templates are not permitted .
Development Period	The project must be developed entirely during the official hackathon period.
AI-Assisted Development	AI tools such as ChatGPT, Claude, GitHub Copilot, Puku, Gemini, and similar tools are permitted.
Open-Source Software	Open-source frameworks, libraries, APIs, SDKs, and publicly available resources may be used. Proper attribution must be provided in the project's README where applicable.
Original Work	The application's architecture, implementation, and business logic should represent the team's own engineering effort.

Submission Requirements

Each team must submit the following before the submission deadline.

Required Submission	Status
Public GitHub Repository	Required
README Documentation	Required
Project Description	Required

Recommended (Optional)

The following items are not mandatory but may strengthen the overall presentation and evaluation of your project.

- Live deployment
- API documentation
- Database schema or ER diagram
- Testing notes
- Setup and installation guide
- Architecture diagram
- Demo video or presentation slides

Important Notes

- The judging panel reserves the right to ask technical questions regarding any part of the submitted project.
- Teams should be prepared to explain their system architecture, AI workflow, implementation decisions, and technology choices during the demonstration.
- Any use of third-party services, APIs, datasets, models, or libraries should be appropriately acknowledged.
- Failure to comply with the competition rules may result in score deductions or disqualification at the discretion of the organizing committee.
- The judges' decision shall be final.

Good Luck!

We look forward to seeing innovative solutions that demonstrate technical excellence, meaningful AI integration, and real-world impact.

- **IEEE Computer Society Southeast University Student Branch Chapter**
- **AI & API Hackathon 2026 – Final Round**